

RAJIV GANDHI UNIVERSITY OF KNOWLEDGE TECHNOLOGIES, ANDHRA PRADESH

RK Valley Institute

(Constituted under the A.P. Govt. Act 18 of 2008 and recognized as per Section 2(f), 12(B) of UGC Act, 1956)

Accredited by NAAC with 'B+' Grade

Department of Computer Science and Engineering

Minor Project under the guidance of

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ABSTRACT

- Mocku is a full-stack platform designed to conduct online mock tests efficiently
- It provides a real-time test interface with automatic evaluation
- Students can track their performance and improve weak areas
- Admin can manage tests, questions, and results dynamically
- Integrated with Google Authentication and secure JWT-based login
- Built using React Native, Node.js, Express, and MongoDB
- Aims to improve exam preparation through structured practice

INTRODUCTION

- Traditional exam preparation lacks structured practice and real-time evaluation.
- Students face difficulty in analyzing performance and identifying weak areas.
- Manual test systems do not provide instant results or detailed feedback.
- Lack of centralized platforms leads to inefficient test management and tracking.
- A digital mock test system is needed to provide real-time practice and performance insights.

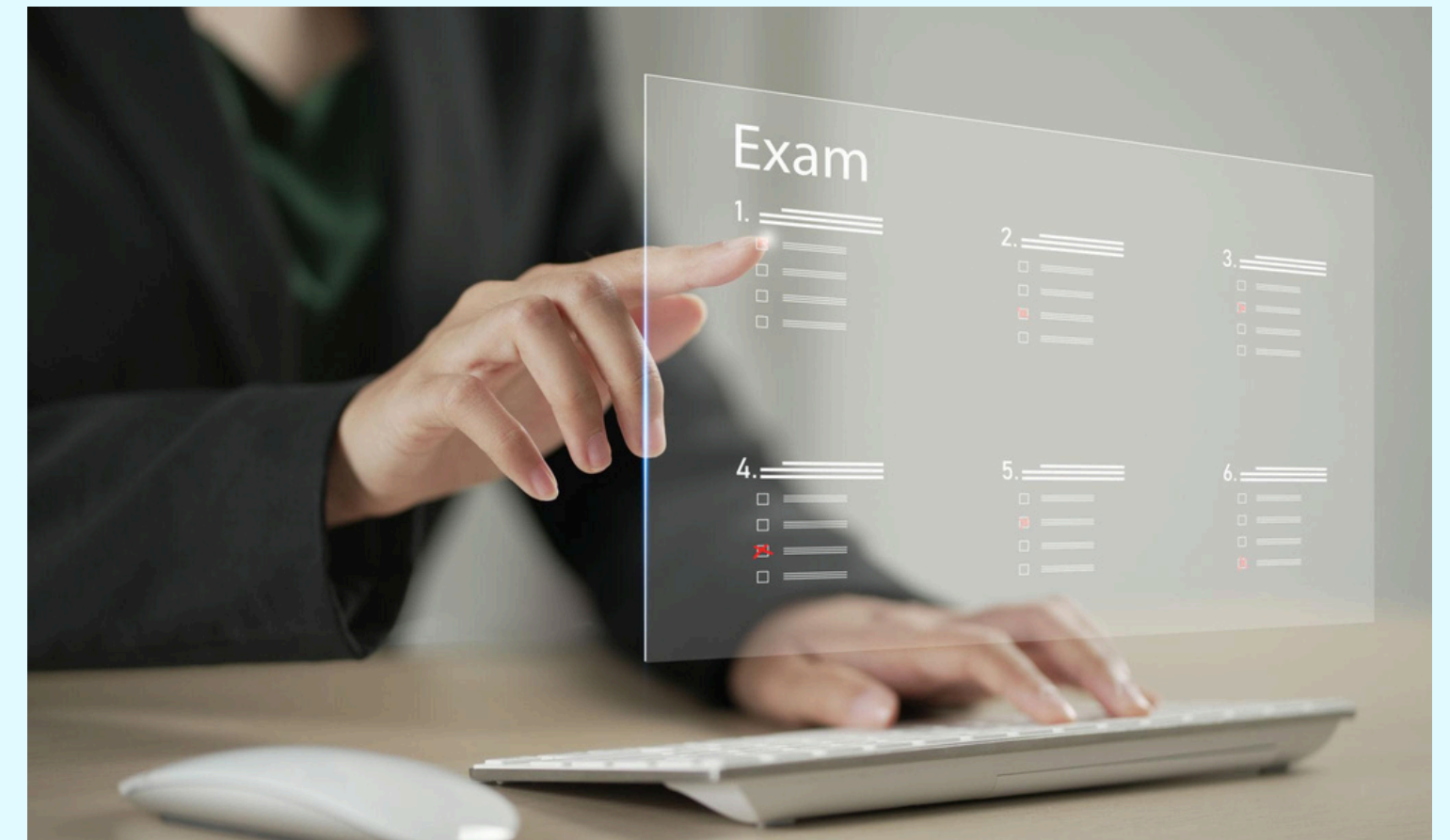
PROJECT OVERVIEW

- A centralized platform connecting students and administrators for conducting mock tests.
- Supports test creation, test attempts, automatic evaluation, and result analysis.
- Provides role-based access for students and admin with secure authentication.
- Ensures efficient data management and real-time interaction between frontend and backend.
- Improves learning efficiency and reduces manual effort in test evaluation.

KEY FEATURES

For Students :

- Registration / Login
- Attempt Mock Tests
- Performance Tracking
- View Results & Analysis



KEY FEATURES

For Admin:

- Create and Manage Tests
- Add / Edit Questions
- View Student Results

For System Admin / Management:

- Manage Test Data
- Monitor Performance Analytics
- Publish Updates / Notifications

SYSTEM ARCHITECTURE

1. User

- React Native (Frontend)
- Node.js & Express (Backend)
- MongoDB Database

2. Protocols: HTTP/HTTPS

SYSTEM MODULES

1. Student Module

- Register / Login to the system
- Attempt mock tests and submit answers
- View results and performance analysis

2. Admin Module

- Create and manage tests
- Add / update questions
- Monitor student performance and results

DATABASE DESIGN

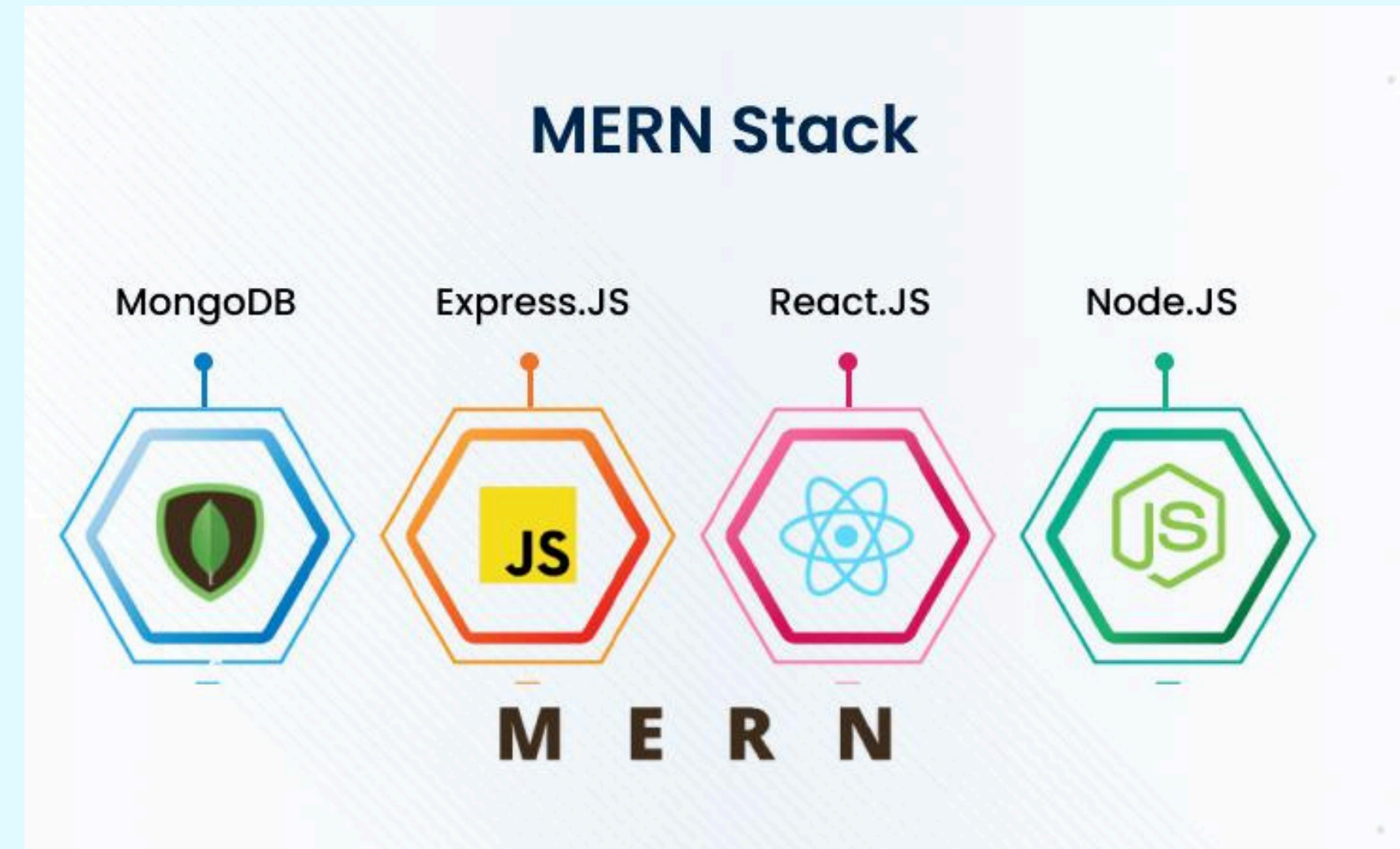
Main Collections:

- Users
- Admin
- Tests
- Questions
- Results
- Exams



TECHNOLOGIES USED

- Frontend: React.JS
- Backend: Node.js & Express
- Database: MongoDB
- Authentication: JWT & Google OAuth
- Tools: VS Code, Postman



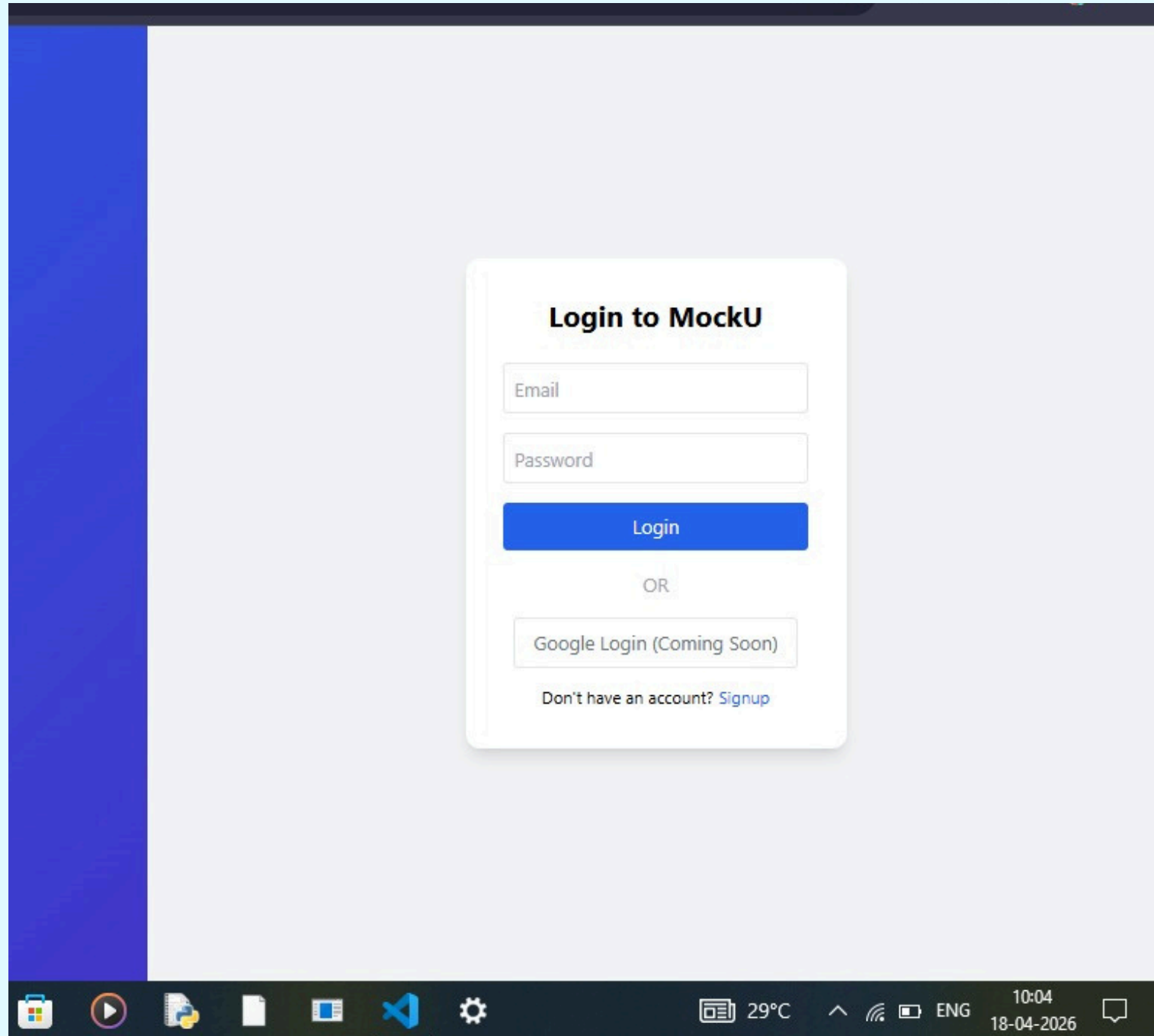
CHALLENGES FACED

- Implementing secure authentication (JWT & Google Login)
- Managing test timer and real-time submission logic
- Handling frontend–backend API integration
- Designing efficient database schema and relationships
- Ensuring smooth and responsive user interface

Work	What It Does	Limitation
Smart Campus Systems	Automates institutional operations	Complex and not flexible for general user needs
RBAC Models	Controls user roles and permissions	Does not handle dynamic user interactions
JWT Authentication	Provides secure login and session handling	Focuses only on security, not full system functionality
MERN Stack Applications	Builds scalable web platforms	Generic systems, not focused on integrated user services
Analytics Dashboards	Tracks and visualizes data	No support for real-time interaction or service management

SCREENSHOTS

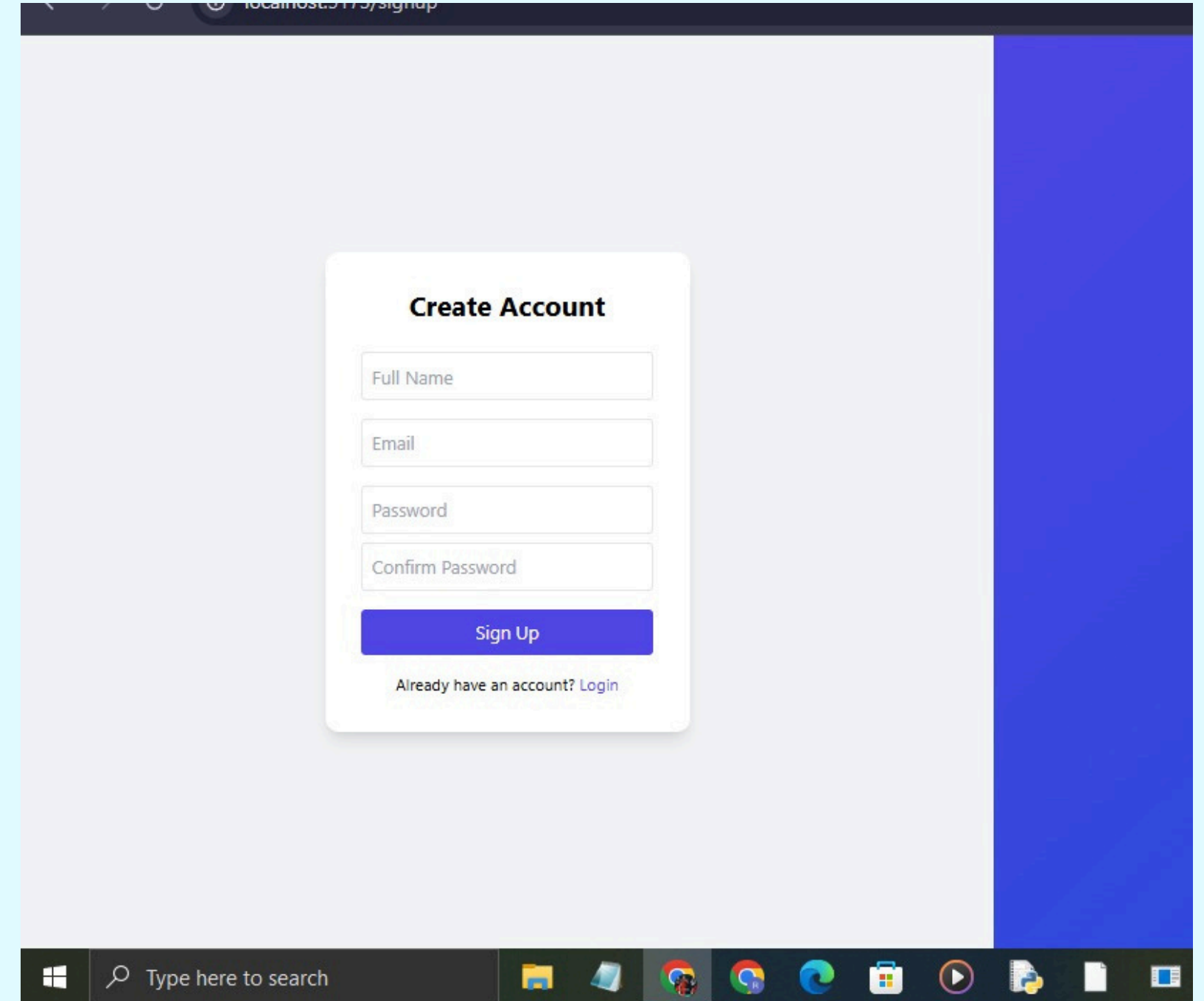
Login Page & Signup Page:



The screenshot shows a web browser window displaying the 'Login to MockU' page. The page has a light purple background with a blue vertical bar on the left. The login form is centered and contains the following elements:

- Login to MockU** (Title)
- (Email)
- (Password)
- (Login Button)
- OR
- (Google Login Button)
- Don't have an account? [Signup](#) (Link)

The Windows taskbar at the bottom shows the time as 10:04 on 18-04-2026, with a temperature of 29°C and various system icons.



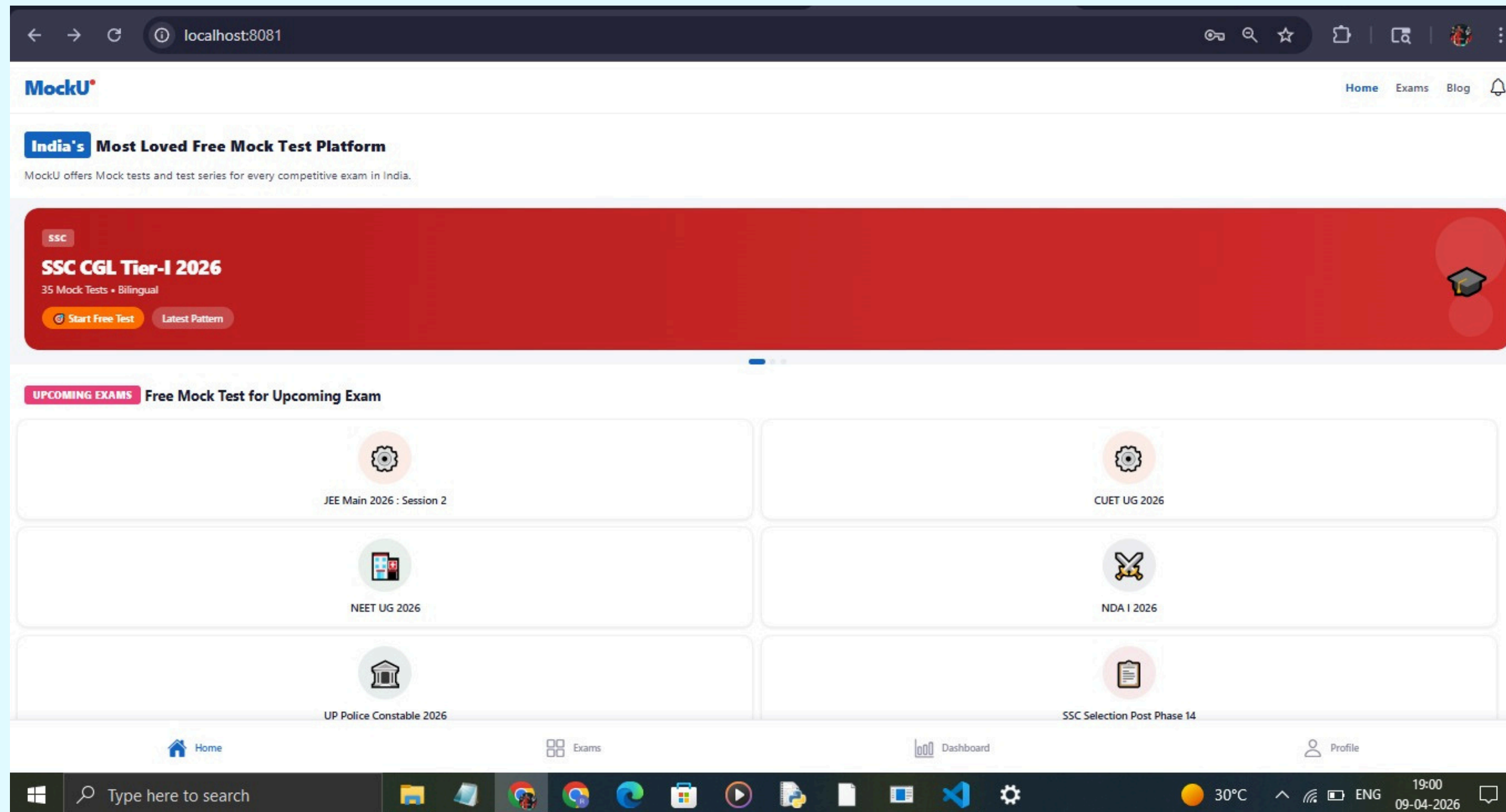
The screenshot shows a web browser window displaying the 'Create Account' page. The page has a light purple background with a blue vertical bar on the right. The create account form is centered and contains the following elements:

- Create Account** (Title)
- (Full Name)
- (Email)
- (Password)
- (Confirm Password)
- (Sign Up Button)
- Already have an account? [Login](#) (Link)

The Windows taskbar at the bottom shows the search bar and various application icons.

SCREENSHOTS

Main Page (Dashboard):



SCREENSHOTS

Test / Exam Section Page:

The screenshot shows a web browser window with the address bar displaying 'localhost:5173/exams'. The page is titled 'Mock Test' and features a timer in the top right corner showing '1:45'. The main content area contains a question: 'Q3. React is a?'. Below the question are four input fields: 'Library', 'Language', 'OS', and 'Browser'. There are three buttons: 'Mark for Review', 'Prev', and 'Next'. A large blue button labeled 'Submit Test' is at the bottom. On the right side, there is a 'Questions' sidebar with three question cards numbered 1, 2, and 3. The Windows taskbar is visible at the bottom, showing the search bar and various application icons.

Mock Test

1:45

Q3. React is a?

Library

Language

OS

Browser

Mark for Review

Prev

Next

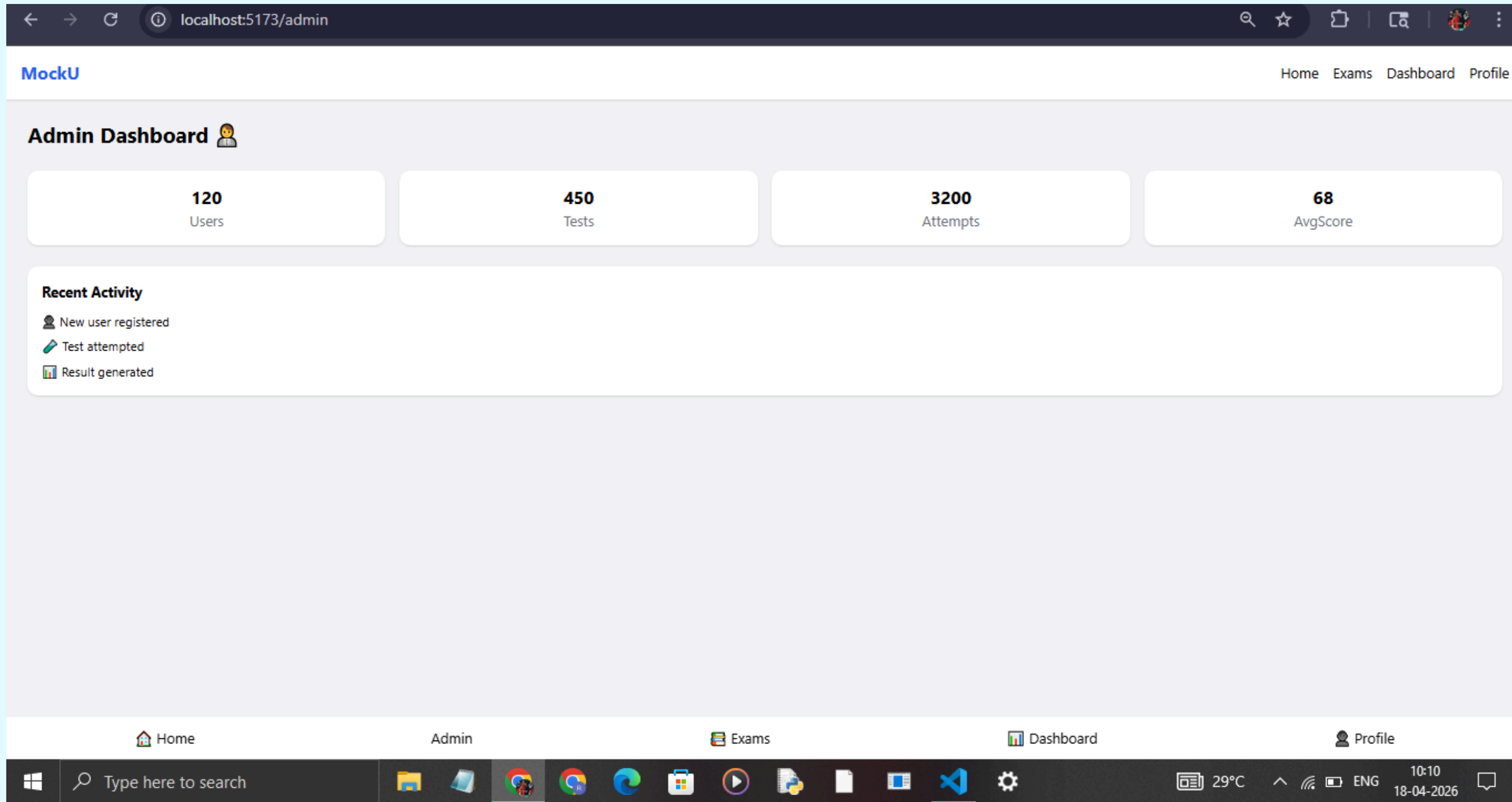
Submit Test

Questions

1 2 3

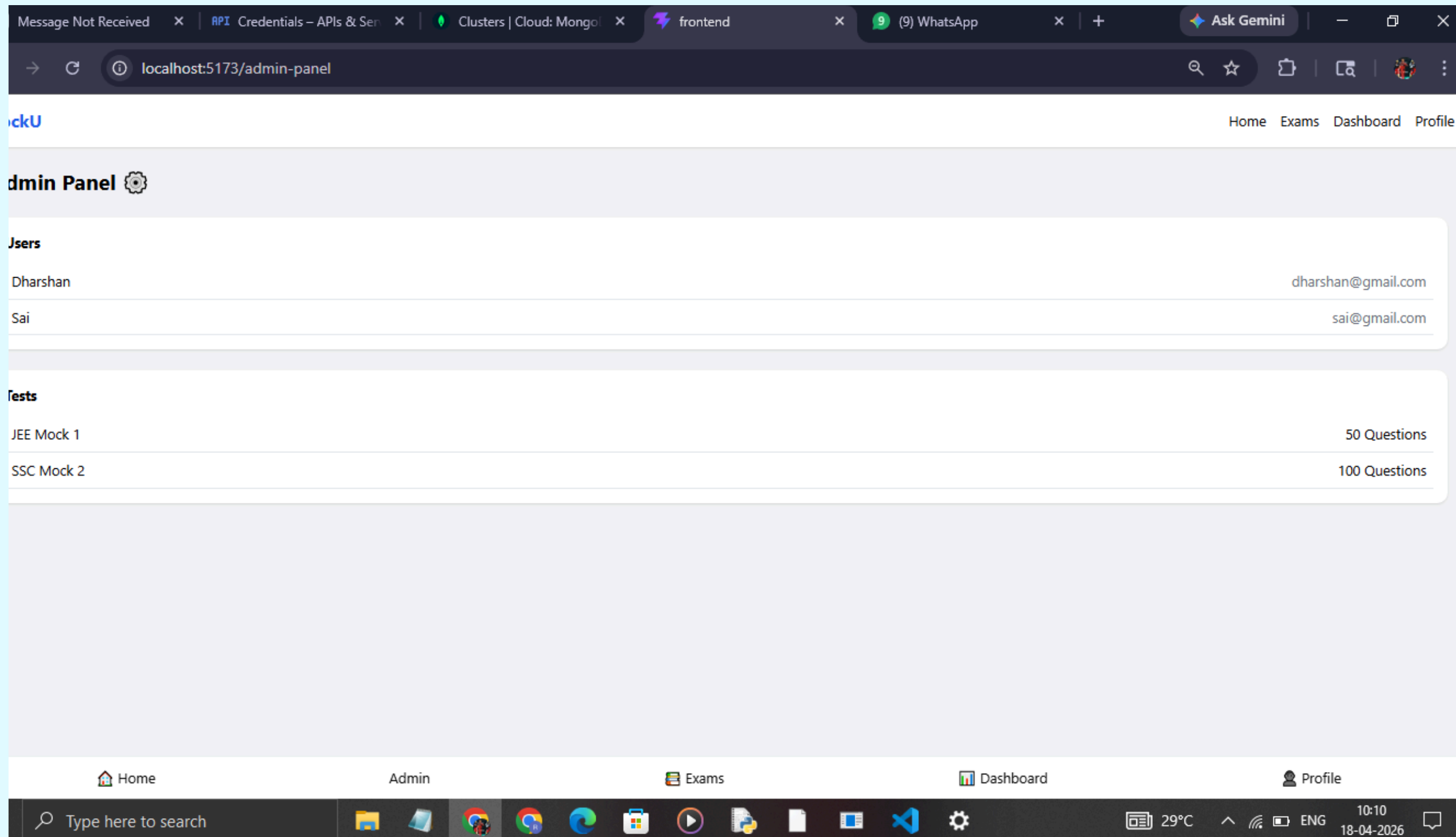
SCREENSHOTS

Admin Dashboard:



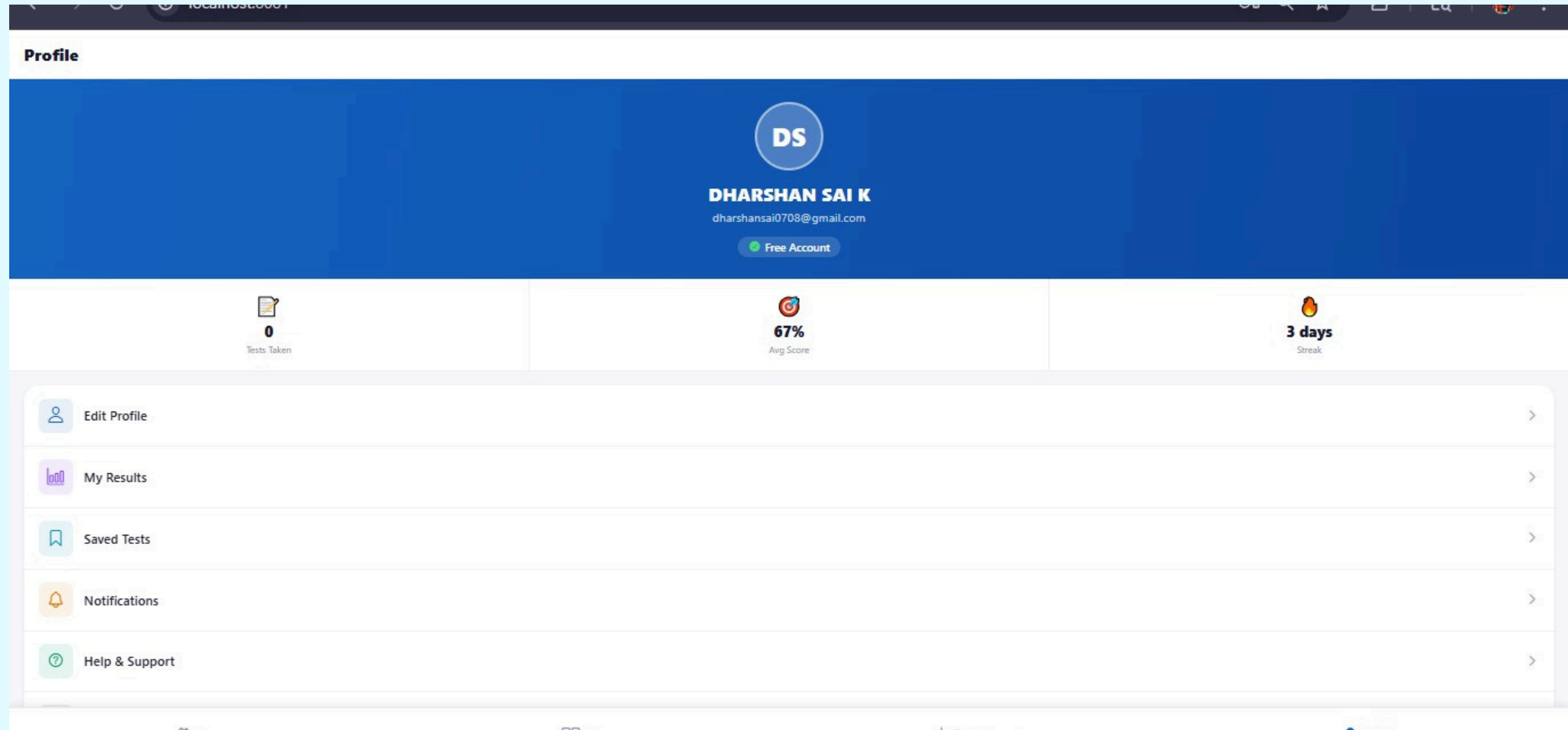
SCREENSHOTS

Admin Functionalities:



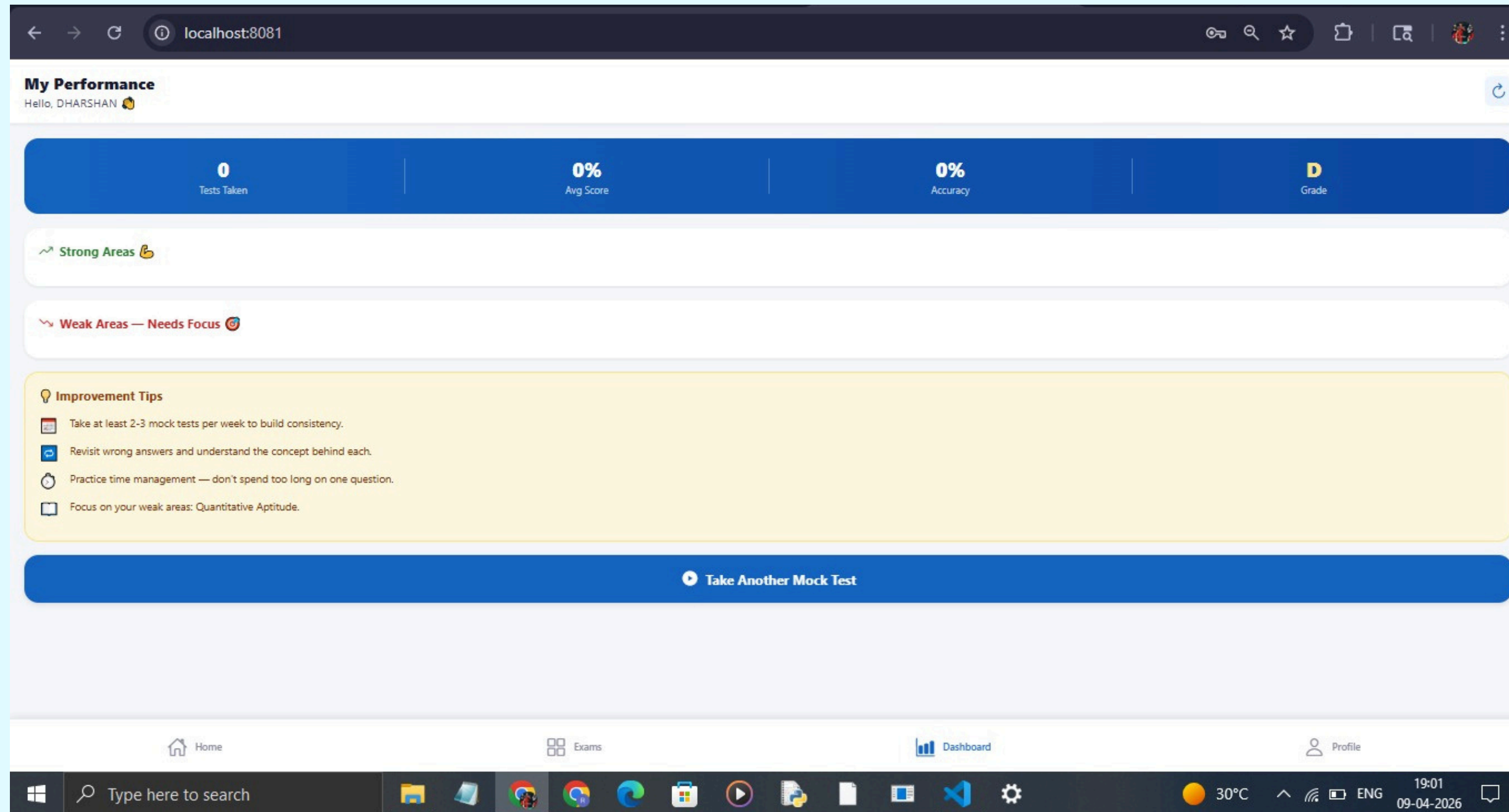
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Student Dashboard:



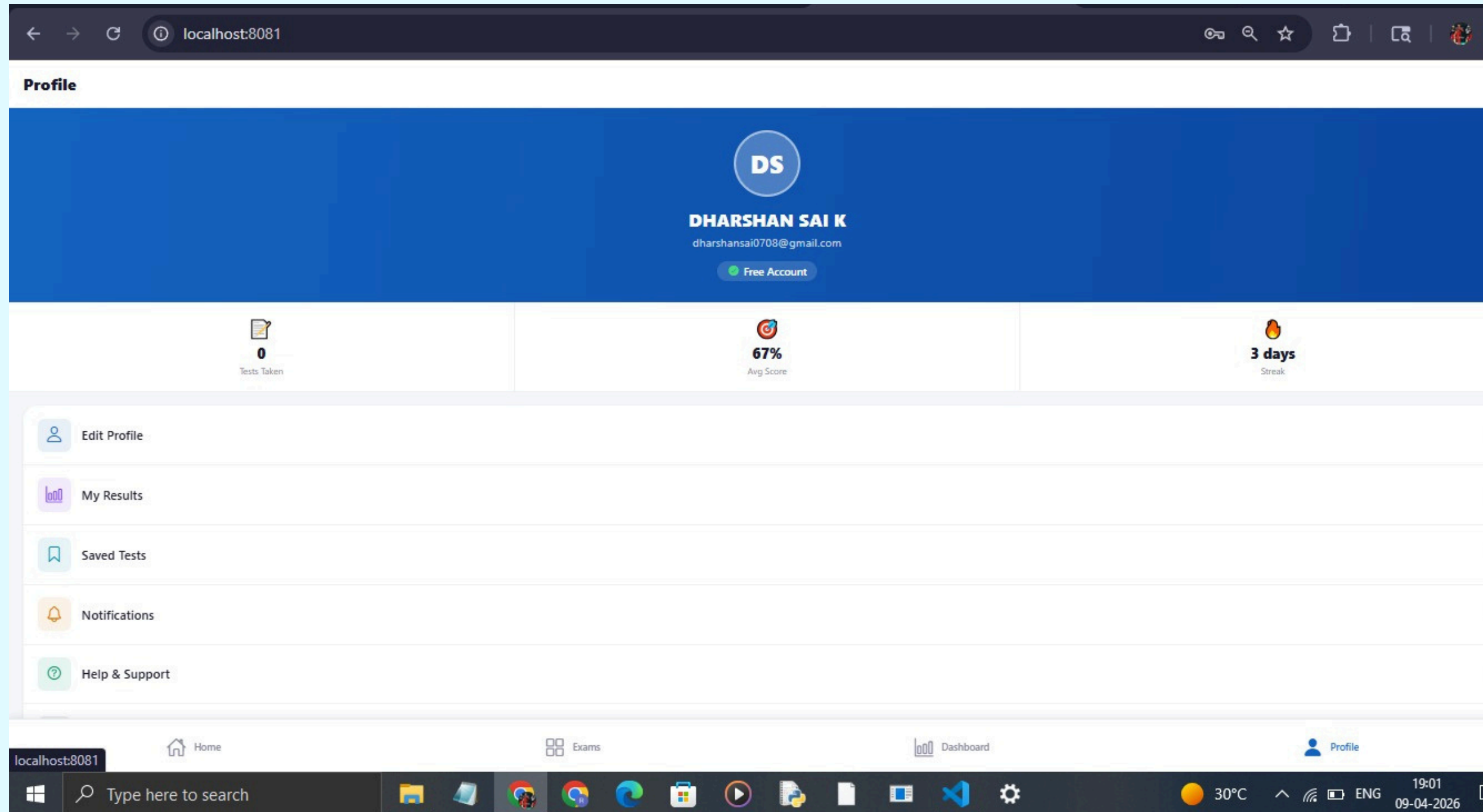
SCREENSHOTS

Student Functionalities:



SCREENSHOTS

Student Functionalities:



FUTURE ENHANCEMENTS

- AI-based performance analysis and personalized recommendations
- Adaptive testing based on user skill level
- Leaderboard and ranking system
- Real-time test analytics and detailed reports
- Mobile application development
- Integration with placement preparation platforms

CONCLUSION

- The system successfully provides an online platform for mock test practice
- Reduces dependency on manual preparation methods
- Enables students to improve performance through regular practice
- Provides instant results and performance analysis
- Scalable for future enhancements and advanced features

REFERENCES

- MongoDB Official Documentation
<https://www.mongodb.com/docs/>
- Node.js Official Documentation
<https://www.nodejs.org/en/docs>
- Expo Documentation
<https://docs.expo.dev/>
- Mockers online website
<https://www.mockers.in/>

THANK YOU